Capacity-Valid Scheduling of a Heterogeneous GPU Cluster

2026 HKUST-GZ MCM Senior Track – Problem B

Main findings: A10 is the main sustained capacity-pressure pool in the immediate-start counterfactual. On the complete A10 trace, capacity-valid SPT-HOL lowers ordinary mean waiting to 32.906 h versus 34.967 h for SPT-FILL, while the fixed GPU-work objective reverses that small ranking. Ratio placement gives zero waiting for all 9,643 GS1 jobs in the measured instance. For substitution, the measured $\sigma = 1$ point favors Hash-10% for SPT-HOL A10 JCT, while Shape-filtered has the lower FCFS JCT at both measured speeds; the choice depends on native harm, completion tails, and donor occupancy.

Data and validation. We use the supplied 224,112-job trace, exact integer-second durations, six model pools, non-preemptive gang starts, and complete drain for waiting/JCT. Allocation rates use the fixed 90-day window. Original IDs and requested fields, node-level CPU/GPU ledgers, full drain, fixed-window counts, and independently recomputed metrics were checked against saved formal schedules. This revision performs no scheduler or replay rerun.

1 Question 1: Workload structure and capacity pressure

The supplied trace has 224,112 jobs on 4,278 nodes, with 10,412 GPUs and 632,636 vCPUs across six GPU models. For job j, a_j is release time, p_j duration, W_j workers, and (g_j, c_j) the per-worker GPU and vCPU requests. Submitted GPU work is $H_j = W_j g_j p_j$; it is a request-based quantity, not measured device utilization. The six model pools have distinct capacities and request shapes; their formal simulated outcomes are reported with the scheduler evidence in Question 2. The exact input hashes and the serialization comparison are in the provenance record, so the claims below use the formal-run CSV.

Table 1: Workload type statistics from the formal raw CSV. Percentages are jobs or submitted GPU-work as labelled; the multi-worker summary reports counts and the median/P95/maximum worker count.

Population	Duration ≤ 1 h (jobs; GPU-work)	Duration > 24 h (jobs; GPU-work)	Worker structure
All 224,112	70.751%; 0.721%	9.295%; 92.350%	$W = 1$: 90.953%; $W > 1$: 20,276 jobs (9.047%). Within multi-worker: $W = 2$–8: 16,559; $W \geq 9$: 3,717; median/P95/max $= 4/28/249$.
A10 111,284	79.035%; 0.176%	10.717%; 99.417%	$W = 1$: 99.293%; $W > 1$: 787 jobs (0.707%). Within multi-worker: $W = 2$–8: 637; $W \geq 9$: 150; median/P95/max $= 4/17.7/60$.

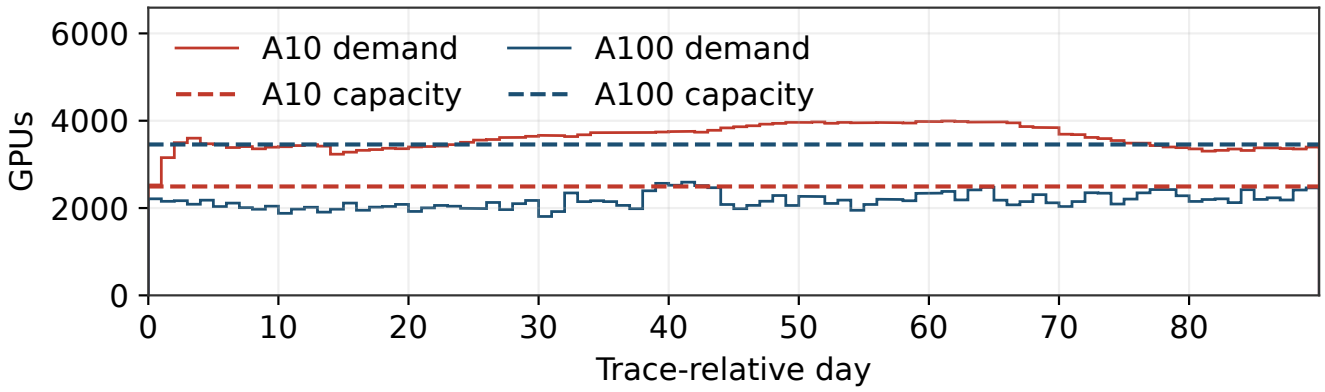


Figure 1: Exact daily-bin overlap of the immediate-start counterfactual and physical GPU capacity. This is demand/capacity, not simulated allocation. The A10 mean is 3,614.54 GPUs against 2,494 capacity and its peak is 4,103; A100 demand stays below its 3,456-GPU capacity.

The workload is concentrated: GPU-work Gini is 0.9522, the top 1% (2,241 jobs) contributes 43.30%, and the top 5% (11,205 jobs) contributes 87.68%. Median duration is 1,330 s, the 95th percentile is 309.09 h, and the maximum is about 184 days. There are 3,512 jobs at trace time zero; without a calendar origin, their position is reported as a trace-origin batch rather than a weekday effect. The exposure-corrected relative 24-hour profile peaks at phase hour 16 and is lowest at hour 4 (ratio 2.29); this is a relative phase diagnostic, not a timezone claim.

The shape is also strongly single-worker: 90.95% of jobs use one worker, 78.42% use one GPU per worker, and the means are 1.539 workers and 2.003 GPUs per worker. Daily arrivals are nonstationary at the trace scale: the first day contains 8,509 jobs, day 1 contains 5,525, and the minimum is 965 jobs on day 63. These counts describe the observed window and do not identify a calendar effect or a particular arrival process.

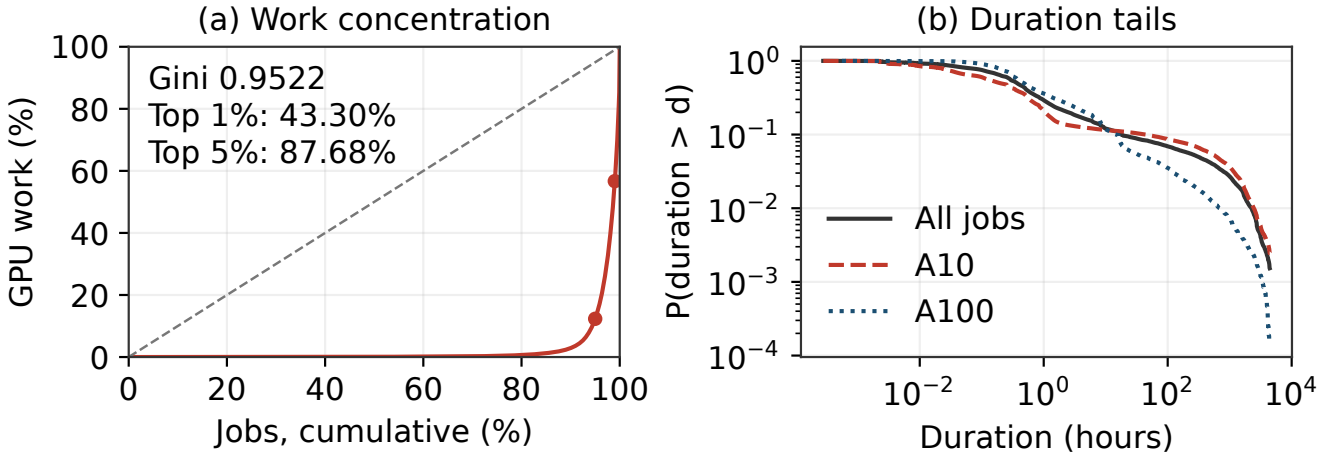


Figure 2: Two high-information views of the formal data: cumulative GPU-work concentration and strict duration tails.

For model m, with exact Q1 observation window $T = 7,775,878$ s, the lifetime-work ratios are

$$\rho_m^{GPU} = \frac{\sum_{j\in\mathcal{J}_m} W_j g_j p_j}{T \sum_{i\in\mathcal{I}_m} G_i}, \qquad \rho_m^{CPU} = \frac{\sum_{j\in\mathcal{J}_m} W_j c_j p_j}{T \sum_{i\in\mathcal{I}_m} C_i}. \tag{1}$$

A10 has $\rho^{GPU} = 2.064$, while GPU-series-1 has a shape mismatch: 2,478 jobs (25.70% of that pool, 78.07% of its work) exceed the 24-vCPU/GPU ratio of its 8-GPU, 192-vCPU nodes. The 24-, 48-, 128- and 192-vCPU/GPU configurations contain 77, 10, 1 and 901 nodes respectively. This motivates node-level gang checks; it does not identify a unique bottleneck policy.

2 Question 2: Capacity-valid scheduling

2.1 Contract and Algorithm 1

Each GPU model is simulated as its own pool. A job starts at $s_j \geq a_j$, completes at $f_j = s_j + p_j$, and all W_j workers start together. At a proposed start time, a placement x_{ji} is admitted only if the *pre-start* residual capacities satisfy

$$g_j x_{ji} \leq G_i^{free}(t), \qquad c_j x_{ji} \leq C_i^{free}(t), \qquad \sum_i x_{ji} = W_j. \tag{2}$$

The resulting full-time model is instead the active-job constraint $\sum_{j:s_j \leq t < f_j} g_j x_{ji} \leq G_i^{cap}$ and its CPU analogue. Thus a single job's check against residual capacity is not written as a whole-cluster dynamic constraint. Durations are exact integer seconds; gang admission is atomic, non-preemptive and non-migrating.

Algorithm 1 (SPT-HOL): For each model pool, insert only arrived jobs into a shortest-processing-time heap keyed by $(p_j, a_j, \mathrm{id}_j)$. Examine the heap head. If it is infeasible on the current residual nodes, stop dispatching this pool at this timestamp (head-of-line blocking). If feasible, scan nodes in the selected placement order, verify every worker slot's GPU and CPU residual, then admit all workers atomically and schedule one completion event. For GPU-series-1 only, the formal C0 case uses a CPU/GPU-ratio node order before the same full slot check; every other pool uses ascending-ID first fit. Advance to the next arrival or completion and repeat.

The historical program label `SJF-HOL` means this same method; the paper name is uniformly `SPT-HOL`. `SPT-FILL` uses the same key but may inspect later feasible jobs under the fixed $K = 200$ bypass budget. A heap insertion or removal costs $O(\log n)$; a dispatch also scans candidate nodes and worker slots, and ratio placement orders the candidate nodes by shape. Multiple starts at one timestamp can repeat these scans, so an $O(\log n)$ event claim would be incomplete. Saved schedules were independently checked by route, physical-capacity and metric validators.

The simultaneous-release mechanism has a precise necessary condition. If f^- is the number of free worker slots immediately before an event, Δf is the number released by jobs finishing at that event, and k is the next job width, admission requires

$$f^- + \Delta f \geq k. \tag{3}$$

Only with a continuously feasible single-worker backlog and zero free slots after the preceding dispatch does this reduce to $\Delta f \geq k$. A finite K candidate budget can leave $f^- > 0$, so the reduction is not generally valid. This is a mechanism explanation for some zero-wait observations, not the unique proved cause of HOL's result and not a starvation-free guarantee.

Table 2: Six-pool SPT-HOL results from formal integrated case C0, whose GPU-series-1 placement is ratio and whose other pools use FF-id. Waiting uses complete drain; allocation and completion use $[0, H)$ with $H = 7,776,000$ s.

Pool	Jobs	GPUs	Mean wait (h)	Median/P99 (h)	GPU/CPU alloc.	90d done
A10	111,284	2,494	32.906	0.023/1542.181	99.922%/17.927%	107,002
A100-SXM4-80GB	80,877	3,456	0.000	0.000/0.000	62.755%/42.032%	80,101
GPU-series-1	9,643	1,558	0.000	0.000/0.000	60.867%/34.814%	8,221
GPU-series-2	13,633	976	0.016	0.000/0.045	56.931%/37.424%	13,606
H800	4,716	1,752	0.016	0.000/0.000	55.816%/52.193%	4,705
A800-SXM4-80GB	3,959	176	0.115	0.000/4.390	57.311%/49.493%	3,943

Table 2 is generated from formal case C0; its exact metadata and ledger paths are recorded in the R2 supplement. The C0 A10 timing is identical by job ID to B2, so changing GS1 placement does not change the A10 values.

2.2 A count lower bound and exact accounting

For the complete A10 set let $C = 2,494$, $G_j = W_j g_j$, $Z(t) = \{j : a_j \le t < a_j + p_j\}$, and let $q(t)$ be the minimum number of jobs removed from $Z(t)$ so that the remaining GPU width is at most C. For a schedule, let $S(t) = \{j : s_j \le t < f_j\}$ and $A(t) = S(t) \cap Z(t)$. At any t, at least $q(t)$ jobs in $Z(t)$ cannot be simultaneously active, hence $|Z(t) \setminus A(t)| \ge q(t)$. Integrating and using the waiting identity gives

$$\sum_j \min(\omega_j, p_j) = \int |Z(t) \setminus A(t)|\, dt \ge \int q(t)\, dt, \qquad \sum_j \omega_j \ge \int q(t)\, dt, \tag{4}$$

so $L_{count} = (3600N)^{-1} \int q(t)dt = 22.3179305700$ h is a lower bound on equal-weight mean waiting. For the observed schedule,

$$\omega_j = \min(\omega_j, p_j) + \max(\omega_j - p_j, 0). \tag{5}$$

The exact full-A10 SPT-HOL JSON accounting is

```
{capacity_lb:22.3179305700, within_window_gap:8.3361103922,
    beyond_duration:2.2524130728, mean:32.9064540350}.
```

These are accounting terms around a relaxation, not causal shares or an exact OPT gap. The conservative certificates $\bar{\omega}/L$ are 1.4744 for ordinary job mean and 1.2090 for GPU-work-weighted mean (the latter compares 677.689933 h with the 560.521368 h weighted bound); they are for this fixed instance and different objectives.

2.3 A10 comparison and worker strata

For the same complete 111,284-job A10 population and fixed native GPU-work weights $H_j = W_j g_j p_j$, the three formal schedules give:

Table 3: Complete A10 waiting comparison. Each row reports ordinary mean, P99, maximum, and the fixed-H_j GPU-work-weighted mean.

Policy (case)	Mean (h)	P99 (h)	Max (h)	H-weighted mean (h)
FCFS (B1)	900.129	1419.616	1423.254	619.352
SPT-HOL (C0)	32.906	1542.181	3017.288	677.690
SPT-FILL (full-A1)	34.967	1575.462	3015.311	677.496

The FCFS, SPT-HOL, and SPT-FILL rows are respectively formal cases B1, C0, and full-A1; the full-H1 schedule supplies the HOL side of the worker strata. Exact repository paths and hashes for these four sources are recorded in the R2 supplement.

SPT-HOL is the average-wait choice on this fixed trace: it lowers ordinary mean waiting by 2.061 h relative to SPT-FILL and lowers P99 by 33.280 h, but its maximum is 1.978 h higher. The management trade-off becomes clearer under the fixed GPU-work objective: the ranking reverses, with SPT-FILL at 677.496 h versus SPT-HOL at 677.690 h. Thus "lower average but a higher extreme" and "different ranking after weighting" are both observed facts, not dominance claims.

Table 4: Formal full-A10 HOL/FILL worker strata. The paired difference is HOL minus FILL; negative means lower HOL waiting. P99 values in the two small strata are reported but marked unstable.

Stratum	Jobs	HOL mean (h)	FILL mean (h)	HOL P99 (h)	FILL P99 (h)	Δ mean (h)	Improve / worsen / same (count)
$W = 1$	110,497	33.125	32.805	1553.319	1539.480	+0.320	54,489/50,566/5,442 †
$W = 2$–8	637	1.529	186.767	15.603†	1952.714†	−185.237	628/8/1
$W \ge 9$	150	4.997	982.722	36.186†	2907.896†	−977.725	149/1/0

Small stratum; P99 is unstable. Source: formal full-H1 and full-A1 schedules.

The strata show why a pooled mean can hide policy preference: HOL is much better for multi-worker jobs, whereas FILL has slightly lower mean and P99 for the dominant single-worker stratum. The paired full-trace totals are 55,266 improved, 50,575 worsened and 5,443 unchanged for HOL minus FILL, with mean difference -2.060576 h. These values are from the same A10 IDs and are not imported from a prototype.

For the fixed GS1 pool, C0's ratio placement is a constructive zero-wait schedule: all 9,643/9,643 jobs have zero waiting and mean waiting 0 h, while the FF-id baselines are 0.029188 h (FCFS, B1) and 0.434277 h (SPT-HOL, B2). Since mean waiting is nonnegative, 0 is the optimum for this fixed GS1 instance. The ratio rule is not generalized to other heterogeneous pools.

2.4 Second-round policy evidence

SPT-HEAD-TIME is a bounded policy with a precise implementation definition. When the SPT head is blocked, the code builds a reservation for the first future release event at which the head has a node-level placement witness. While that reservation is active, a backfill candidate is admitted only when it is currently feasible and its finish time is no later than the promised release; a candidate that crosses the promise is rejected by the TIME rule. The $K = 200$ budget counts candidate attempts only after the blocked head has started a bypass scan. Under the shared-timer event mode, timer thresholds at each arrival plus θ create additional dispatch events; for this policy they do not mark jobs urgent. These statements follow the implementation recorded in the R2 supplement, which gives the exact source lines and saved summary.

On the full A10 trace, this second-round case (full-C1, $\theta = 168$ h, $K = 200$, shared timer) changes mean waiting from 32.906454 to 32.923817 h, P99 from 1542.181042 to 1541.586592 h, and maximum from 3017.288056 to 3017.727778 h; its higher cost is not hidden by a single tail statistic. The K/timer and HEAD-TIME results support a bounded mechanism interpretation, not starvation freedom or a general performance theorem. The urgent policies remain prefix-only and are not promoted to the complete-trace recommendation.

3 Question 3: Fixed-split cross-GPU substitution

3.1 Model and two route rules

A job has native model n_j, actual model m_j, duration p_j, workers W_j, and per-worker requests (g_j, c_j). For the sole assumed foreign edge A10→A100-SXM4-80GB, $p'_j = \sigma p_j$; compatibility and σ are scenario assumptions, not measured hardware facts. All workers start together on physical nodes, with no migration or preemption, and every start passes the same CPU/GPU slot check.

The no-substitution baseline keeps each job in its native pool. The Hash-10% rule reserves 60 of 432 A100 nodes for a deterministic 10% selected A10 subset. The implemented Shape-filtered rule keeps only selected jobs whose CPU/GPU shape fits the donor ratio, yielding 10,604 routed and 605 shape-rejected A10 jobs. It is a fixed physical counterfactual, not a claim that a dynamic policy has been implemented. The formal source mapping to the corresponding C0–C4 cases and historical sensitivity cases is in the supplement.

Table 5: SPT-HOL fixed-split evidence from the formal C0–C4 cases. Native harm is the number of native A100 jobs with positive additional waiting relative to No substitution; the final column is the principal harmed A10 group relative to that baseline.

Scenario	σ	A10 wait (h)	A10 JCT (h)	All JCT (h)	90d done	Native harm / A10 harmed
No substitution	–	32.906	132.676	93.353	217,570	0 / 0
Hash-10%	1	20.534	120.303	87.209	217,814	0 / 49,772
Hash-10%	2	25.106	134.810	94.412	217,521	0 / 56,422
Shape-filtered	1	24.359	124.128	89.108	217,801	0 / 46,803
Shape-filtered	2	24.359	129.671	91.860	217,613	0 / 55,465

For FCFS at the measured $\sigma = 1$ point, Hash-10% gives A10 mean wait 587.875 h and all-job mean JCT 368.910 h; Shape-filtered gives 656.356 h and 402.914 h. At $\sigma = 2$, FCFS Hash-10% has A10 mean JCT 831.456 h, whereas Shape-filtered has 761.668 h. Thus Shape-filtered has the lower FCFS JCT at both measured speeds, while Hash-10% is better for SPT-HOL at the measured $\sigma = 1$ point (A10 JCT 120.303 versus 124.128 h). These are measured points, not an upper bound over an interval, an unknown-speed guarantee, or a unique σ^*.

3.2 What the trade-off means

For SPT-HOL, Hash-10% is the more aggressive 10% split; Shape-filtered rejects 605 high-CPU/GPU jobs and changes the selected load. At the measured $\sigma = 1$ point, Hash-10% reduces A10 wait relative to No substitution, while Shape-filtered has a smaller A10 benefit but lower paired positive completion-delay mass (16,051.659 job-hours versus Hash-10% at 68,183.183 job-hours). At $\sigma = 2$, common routed-job ordering is unchanged because all durations in one foreign queue are multiplied by the same σ; re-simulation is still required because completion/release interactions change. Shape-filtered's A10 wait is unchanged at 24.359 h between the two measured speeds, but A10 JCT rises 5.542 h and 90-day completions fall by 188.

The A100 physical allocation comparison uses one full-pool denominator of 3,456 GPUs and sums native plus foreign occupancy. Hash-10% at $\sigma = 1$ is 0.7224 (native 2,976-GPU partition 0.6276 plus foreign 480-GPU partition 0.0948). Shape-filtered is 0.6899 at $\sigma = 1$ (0.7288 native plus 0.4492 foreign) and 0.7148 at $\sigma = 2$ (0.7288 plus 0.6284). Thus the foreign partition is 44.923% allocated and 55.077% idle at $\sigma = 1$, versus 62.841% allocated and 37.159% idle at $\sigma = 2$; neither fraction is carried across scenarios. A larger partition rate is not called a utilization gain by shrinking the native denominator.

The aggregate capacity conversion $534/8 = 66.75$ is only a pooled arithmetic quantity; it is not rounded to 67 nodes and is not a node-level safe-borrow certificate. The conclusion that 60 donor nodes produce zero native harm is supported by the formal physical schedule, while host safety still depends on node-level placement checks.

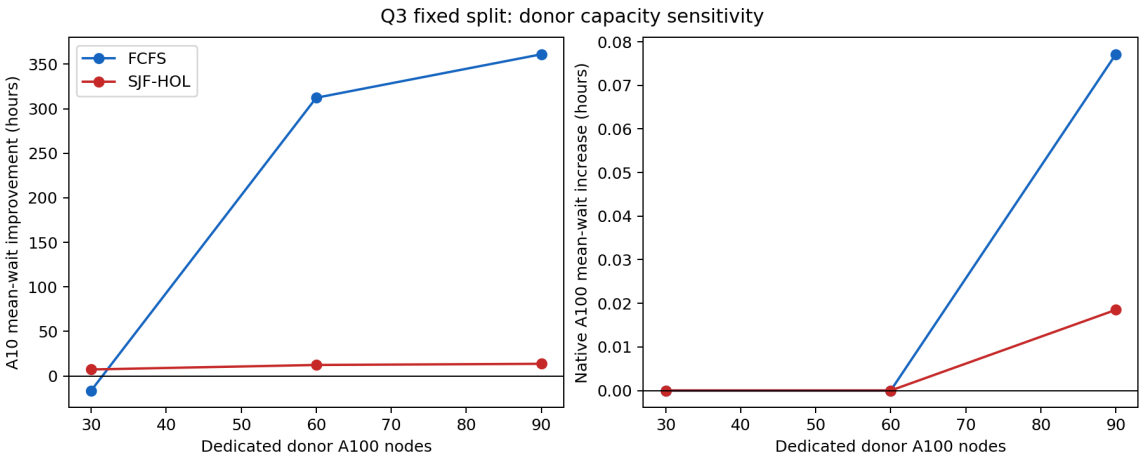


Figure 3: Historical donor-capacity sensitivity. The left panel is A10 mean-wait change; the right panel is native A100 mean-wait change, with separate scales.

The historical sensitivity retains its negative cases: at 60 donors, 20% routing raises FCFS A10 mean wait to 1,010.136 h and its maximum to 5,445.367 h; 30 donors at 10% gives 916.407 h, worse than No substitution. For the 90-donor case, SPT-HOL native A100 has 2,150 positive-delay jobs, mean additional wait 0.018492 h over all 80,877 native jobs, and harmed-subgroup mean 0.695606 h. These finite paired results expose the management choice between A10 relief, native harm, tail, and completion count.

A financial extension is optional rather than a second experiment. If a tenant values completion time at v per hour and pays P, voluntary substitution requires payment reduction at least v times the completion-time increase. No price, participation rate, revenue, or profitability is inferred here. An online completion-cost routing rule is future work; no additional batch policy or borrowed threshold is introduced.

4 Conclusion, strengths, limits, and provenance

The three questions answer different decisions. Q1 separates submitted work, immediate-start demand, and simulated allocation. Q2 provides a capacity-valid SPT-HOL contract, a count lower bound, a zero-wait GS1 construction, and complete-A10 tail and cost evidence. Q3 compares fixed physical partitions and measured speed assumptions while retaining native harm and duration sensitivity. These are finite trace results: allocation is request occupancy, GPU-hours across models are not interchangeable hardware or monetary units, and no heuristic optimality, speed guarantee, or dynamic substitution policy is claimed. An online completion-cost routing rule remains future work.

Strengths and limitations. Strengths are exact integer-second accounting, node-level CPU/GPU gang checks, saved-ledger replay, and separate ordinary and GPU-work objectives. Limitations are the finite trace, request-based occupancy rather than device telemetry, unbenchmarked cross-GPU compatibility and speed assumptions, and fixed-policy rather than dynamic substitution. The reported choices are conditional on the measured cases and do not establish a universal optimum.

References

1. HKUST-GZ MCM 2026 Senior Track, Problem B, repository copy (official question PDF; hash in the source index).
2. Graham, R. L. (1966), "Bounds on multiprocessing timing anomalies," *SIAM Journal on Applied Mathematics*, 14(2), 416–429. Used only as scheduling background; the reported lower bound is derived for this trace.
3. Feitelson, D. G. (2015), *Workload Modeling for Computer Systems Performance Evaluation*. Used for workload-trace terminology and the distinction between demand and allocation.
4. SchedMD, "Backfill scheduling" documentation. Used to distinguish a feasible-fill bypass from SPT head-of-line blocking.

Use of AI Tools

ChatGPT and OpenAI Codex were used throughout the research for discussion, source and data inspection, mathematical-notation assistance, read-only post-processing, plot generation, code and evidence cross-review, and English drafting/typesetting. Claude also provided substantive assistance in the research record: candidate lower-bound and decomposition reasoning, ratio-placement analysis, prototype analysis, and independent cross-review. Formal scheduler runs, saved-ledger replays, all numerical recomputation in this paper, and final submission decisions were carried out or checked by the team; prototype suggestions were treated as hypotheses and were not substituted for formal evidence. The team remains responsible for the final wording, identity information, and human review.